CS F364: Design & Analysis of Algorithms

Quiz 3 — June 9, 2026

Coverage: Lectures 1–12

> **Note**
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> **Instructions:**
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> - Time: 15 minutes.
> - Write your answers clearly on paper, scan, and upload to Google Classroom.
> - Show all steps for full credit.

Question: Median of Medians

The **Deterministic Select** algorithm (Median of Medians) finds the i^{th} smallest element in $O(n)$ time by:

1. Dividing the n elements into groups of **5**.
2. Finding the median of each group (by brute force, $O(1)$ per group).
3. Recursively finding the median x of these $\lceil n/5 \rceil$ medians.
4. Partitioning around x and recursing on the appropriate side.

The recurrence for groups of size **5** is:

$$T(n) \leq T\left(\frac{n}{5}\right) + T\left(\frac{7n}{10}\right) + O(n)$$

(a) **[2 marks]** Give the recurrence $T_3(n)$ for the Deterministic Select algorithm when groups of size **3** are used instead of 5. What fraction of elements is guaranteed to be $\leq$ the median-of-medians, and what fraction is guaranteed to be $\geq$ it?

(b) **[2 marks]** Give the recurrence $T_7(n)$ when groups of size **7** are used. What are the corresponding fractions?

(c) **[1 mark]** Solve $T_3(n)$ using the substitution method (guess $T_3(n) \leq cn$ and show it fails, then state the actual asymptotic growth).